



Real-Time Simulation as Operational Validation Means of the European DAA System

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This paper presents the methodology, execution, and results of the Real-Time Simulation (RTS) campaign conducted within the European Detect and Avoid System (EUDAAS) project, focusing on the operational validation of Detect and Avoid (DAA) systems with both Remote Pilots (RP) and Air Traffic Controllers (ATCO) in the loop. The EUDAAS project aims to develop and validate a European DAA solution to safely integrate Remotely Piloted Aircraft Systems (RPAS) into non-segregated airspace. The RTS activities were carried out using a detailed RPAS simulation facility equipped with the DAA model and focused on evaluating both Remain Well Clear and Collision Avoidance functionalities for MALE and Tactical UAV platforms. Human factors were central to the validation process, with emphasis on RP and ATCO perceptions of safety, workload, and system usability. Performance data were collected through system recordings, questionnaires, and interviews. Results confirm the operational feasibility of the EUDAAS DAA concept and provide key insights into user interaction and acceptability in both nominal and non-nominal scenarios. ATCO rated the system's usability and acceptability positively, due to limited impact on their operational tasks, while RP expressed lower ratings, mainly due some deficiencies that limited full evaluation of the DAA concept. Nevertheless, all participants agreed on the necessity of a DAA system for the effective integration of RPAS into the civil airspace and the findings led to several recommendations that will support the refinement of the future European DAA concept and its system design.

I. Introduction

The paper describes the activities and results carried out in support of the Operational Validation of the EUDAAS DAA Systems by means of Real-Time Simulation with pilot and controller in the loop.

The European Detect and Avoid System (EUDAAS) project aims to develop and validate, through simulations and flight tests, a European Detect and Avoid (DAA) solution as well as increase the maturity of non-cooperative sensors, to enable the use of Remotely Piloted Aircraft Systems (RPAS) in a much wider and flexible way than currently possible. DAA will enable safe integration of RPAS into non-segregated airspace so that RPAS can operate along with other manned and unmanned aircraft, in order to train for, and conduct, security and defence missions in

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peacetime, crisis and conflict. The project addresses the current user needs by focusing on specific use cases for two types of RPAS: Medium Altitude Long Endurance (MALE) and Tactical UAV (TUAV).

The operational validation through Real-Time Simulations (RTS) has been structured in RTS with only the Remote Pilot in the loop, and RTS activities with both Remote Pilot (RP) and Air Traffic Controllers (ATCO). This paper focuses on the simulation campaign organization, execution and results of this second set of RTS.

These RT simulations have been performed using the simulation environments with integrated DAA system model developed in preceding work packages and evaluate both Remain Well Clear (RWC) and Collision Avoidance (CA) capabilities reproducing detailed dynamic behaviour of MALE and TUAV host RPA's.

The RT simulations aim to measure performance in relevant key areas applicable to both Remote Pilot and ATCO. The Real-time test plan involves highly experienced professional Remote Pilot and ATCO. Specific tools have been used for the recording of data, video and audio, as required for the performance evaluations as well as specific human factors tools such as questionnaires and interviews.

The paper is structured with the following contents. After the present introduction, the second section provides a synthetic recap of the DAA Operational Validation strategy together with the list of applicable validation objectives. The third section focuses on a description of the high-level functional architecture of the available RTS facilities. Moreover, this section describes in detail the attributes, with relative quantitative ranges, of the reference scenarios and of the use cases elements, which are implemented by using the RTS infrastructure described in the previous section.

A fourth section describes the approach, the methods and the procedures that are used to perform Human Factors Assessment. The two final sections address, respectively, the actual execution of the real-time simulations, as carried out at CIRA premises, involving Remote Pilots and Air Traffic Controllers in the validation activities of the EUDAAS system, and the results of the Human Factors Assessment performed by ONERA.

II. Validation Strategy and Objectives

The objective of the Operational Real Time Simulations with human in the loop is to validate a subset of the DAA system requirements in a realistic operational environment. In particular, it aims to provide evidence of the acceptability of the proposed DAA concept by the relevant actors, pilots and air traffic controllers.

The adopted validation strategy is based on the use of a "modelling" method. In particular it consists in integrating a DAA simulation model within a real-time human in the loop simulation infrastructure, able to reproduce, as much as possible, the context (i.e. outside world conditions) in which the RPAS equipped with the DAA system is normally operated.

The context, or outside world includes a list of HW/SW components aimed to simulate:

- the reference scenario, specified in terms of traffic conditions, Air Traffic Management (ATM) rules and procedures, configuration and status of the voice link between ATCOs and pilot's (latency and disturbances), weather conditions and so on;
- the use cases, describing the specific encounter conditions finalized to the activation of the DAA system, specified in terms of encounter geometry and relative speed, intruder type and equipment.

The validation objectives are written by using a language that facilitates the specification of the RT test cards in terms of applicable use cases and reference scenarios. They can be classified in different classes related for example to the System Performance, the System Safety, and the Human Performance and Acceptability.

In our case they will be defined in line with the aim to give evidence of the acceptability by the relevant actors of the following operational aspects:

- Remote Pilot acceptance for DAA timing of alerts
- Remote Pilot acceptance for DAA guidance indications both from RWC and CA
- Remote Pilot acceptance and correct understanding of the DAA Human Machine Interface (HMI)
- Quantification and ATCO acceptance of workload required to manage RPA equipped with the DAAS
- ATCO acceptance of the level of overlap between alerts timing of DAA and ATC ground safety nets, in particular with STCA (Short Term Conflict Alert) [1]
- ATCO and RP acceptance of the perceived level of safety during encounter in nominal conditions
- ATCO and RP acceptance of the perceived level of safety during encounter in non-nominal conditions

By considering only the overall project validation strategy we may assume as first attempt that the applicable Validation Objectives are those related to the evaluation of the Human Performance aspects.

III. RTS facility High-Level Architecture and Reference Scenarios

The block diagram below, Fig. 1, describes the high-level functional architecture of the simulation facility which has been used to perform the DAAS validation by means of RTS human in the loop simulations.

Some relevant elements of the reference scenarios are presented in the following paragraphs.

The test shall cover the airspace classes from A to E (controlled airspace). No test will apply to class G airspace. In Europe, in almost all the European Union countries, the upper airspace is Class C, while the Class A airspace is limited in few countries to Major TMA.

The RPAS will operate under IFR conditions.

The RTSs refer to the presence of both RP and ATCO and to their interactions with the DAA System (DAAS). Thus, it is considered that the Human Factors assessment mainly refers to the effects that the DAAS could have on the coordination between pilots and controllers, on the delivery of clearances by the ATCO and on the compatibility between the DAAS alerts and the TCAS-II advisories.

Different values of traffic density and communications delays also have to be compared for the effect they can have on ATCO-RP interactions.

The MALE and TUAV performances shall be considered in defining the test matrix.

In order to identify the encounter geometries, different sources and previous experiences have been used, that finally supported the selection of encounter geometries specifically addressing the different airspace classes under investigation.

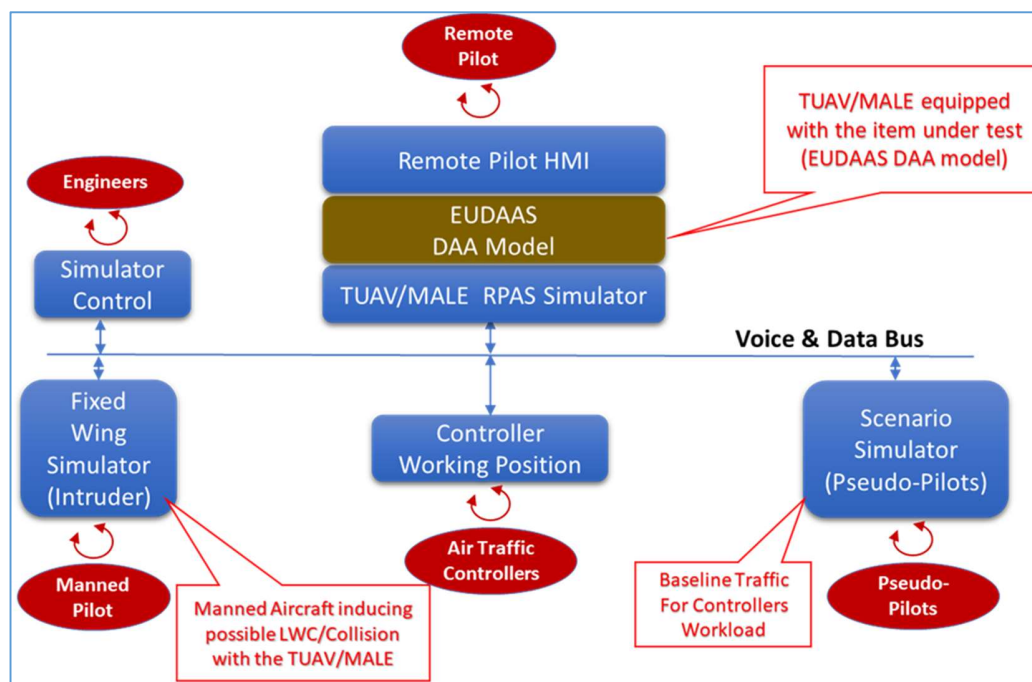


Fig. 1 High-level functional architecture of the real-time simulation facility.

The following specific test conditions, other than nominal ones, have been identified:

- High Latency of the C2 link
- Loss of the C2 link
- TCAS-II compatibility
- DAA Nuisance, here intended as False Alarm condition
- Latency and/or disturbances over the pilot/controller voice communication channel

A. EUDAAS Integration on CIRA RPAS Simulator

The RPAS simulator of the CIRA Integrated Simulation Facility (ISF) is a full detailed dynamics simulator of an RPAS, fully reconfigurable and performance are completely customizable, [2]. Using as references the performance map provided by the EUDAAS project both for TUAV and MALE realization, the following performances characterized the RPAS behavior in the two tested configurations.



Fig. 2 CIRA RPAS (TUAV and MALE) Human Machine Interface.

The CIRA RPAS simulator also implements a fully functional Remote Pilot Station, Fig. 2, that allows for direct pilot control of the RPA, as also the command through autopilots commands and an automatic Flight Management System (FMS) for the autonomous execution of pre-charged flight plans.

The systems integration of the CIRA RPAS and EUDAAS system has been realized as described in the Fig. 3.

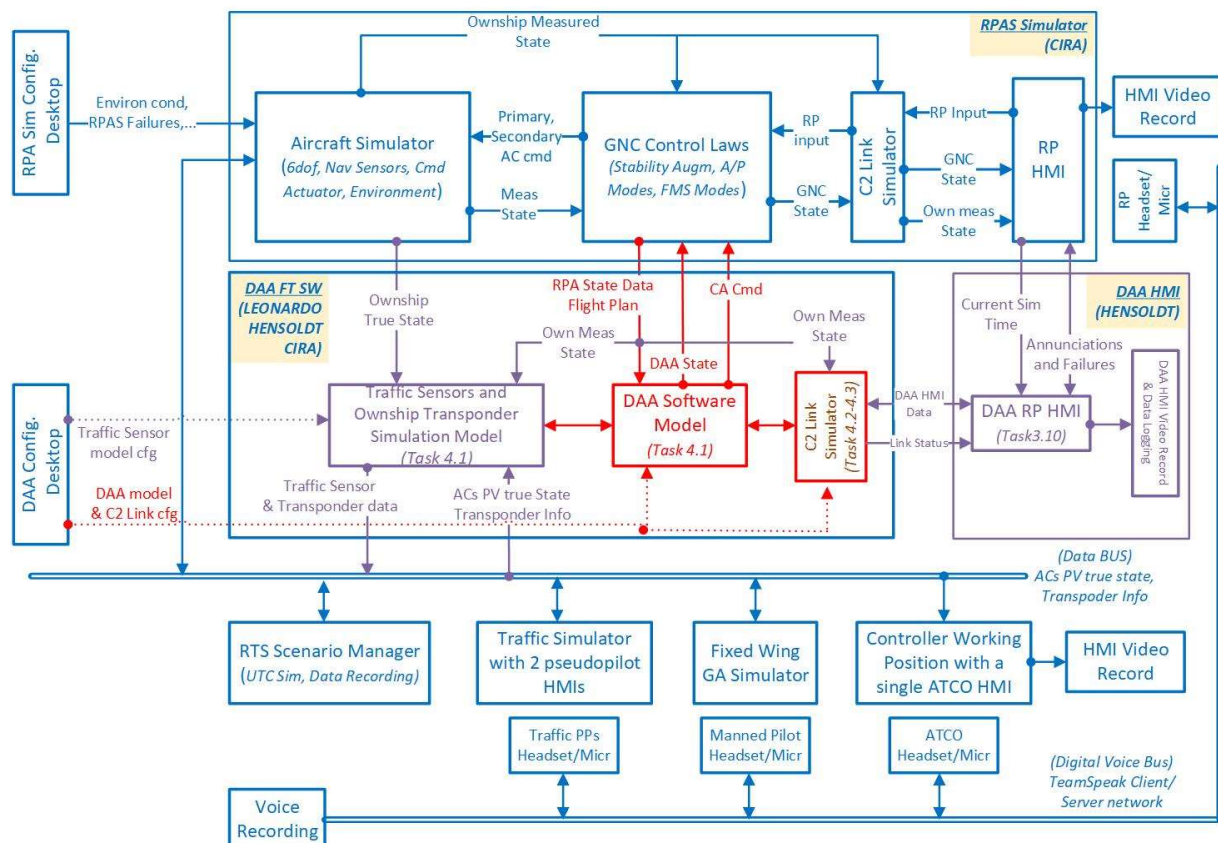


Fig. 3 Architecture of EUDAAS system integration into CIRA RPAS simulation facility

As described in the Fig. 4, the DAA HMI is implemented in a separated HW, namely a notebook that has been placed close to the RPAS-HMI, following the RPs preferences and indications.



Fig. 4 RPAS HMI and left side allocation of EUDAAS HMI.

B. Traffic Simulation

All the simulated scenarios for RTS with RP and ATCO in the Loop include air traffic, with the aim to simulate a workload level for ATCO as close as possible to actual conditions of operations.

The implemented traffic could span among different EUROCONTORL BADA classes of performance [3], which have been selected on the base of airspace class and type of encounter under test.

Table 1 Traffic aircraft performance envelopes (TAS vs. FL).

BADA Class	Description	Full envelope Max FL	Full Envelope Speed Range (Kts)	Speed Range @FL200
1	Piston VLA, small GA	185	70-220	100-220
1	Piston VLA, small GA	185	70-220	100-220
2	Turboprop GA up to 5 seats	285	110-300	150-300
2	Turboprop GA up to 5 seats	285	110-300	150-300
3	Turboprop Airliners (ATR42-72)	415	110-340	160-310
4	GA Business jet up to 15-20 seats	415	120-460	170-380
5	Medium Jet Airliner (B737, A320)	660	135-470	210-400

The Traffic aircraft will be managed by the ATCO and controlled by the pseudo-pilot, as appropriate.

The list of commands applicable by the pseudo pilot are exemplified in the Fig. 5 The behavior of the traffic aircraft has been verified in preliminary tests with RP and ATCO and considered as appropriate for the RTS and its objectives.

As specifically required by EUDAAS project, all traffic aircraft can set emergency codes to be broadcasted to ATCO and RPAS RP, and specifically the code 7400 for VHF Voice Communication channel issues.

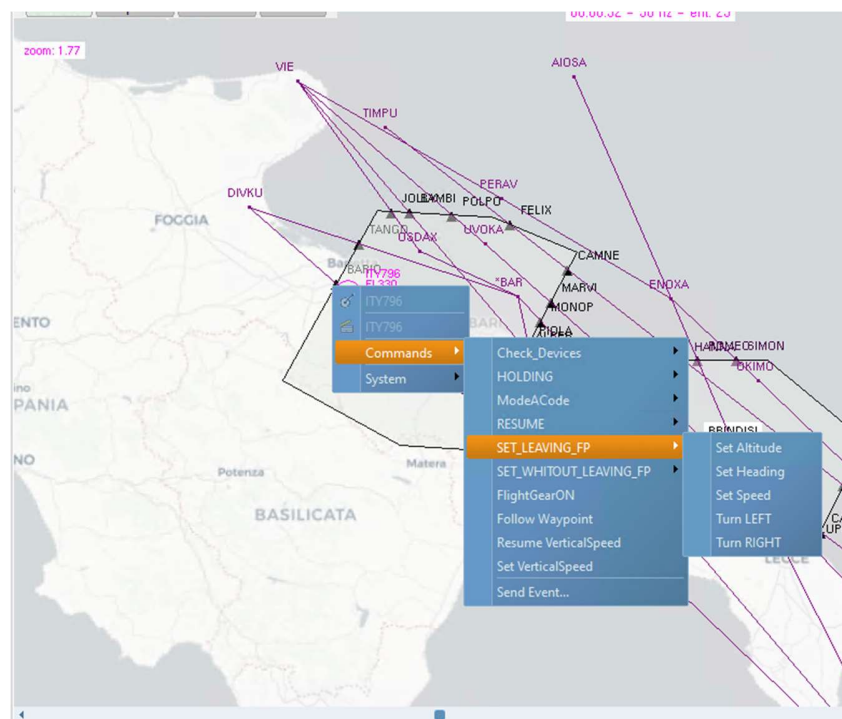


Fig. 5 List of applicable traffic aircraft commands

Four basic traffic scenarios have been designed to support RTS, depending on the airspace class and the kind of operations under test.

The RTS with ATCO are expected to be simulated in airspace classes A, C, D and E. Operations in classes A and E are addressed in the BRINDISI-CTR sector, and operations in classes C and D are addressed in the BARI-CTR sector. The following reference scenarios for traffic have been used, see Fig. 6 to Fig. 9.

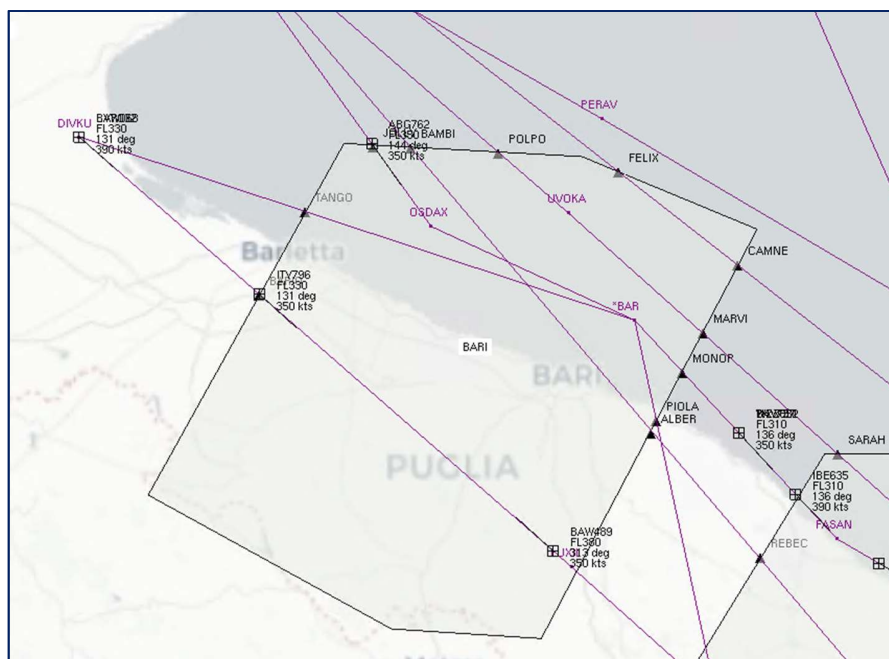


Fig. 6 Airways and SIDs/STARs used for operations in airspace classes C

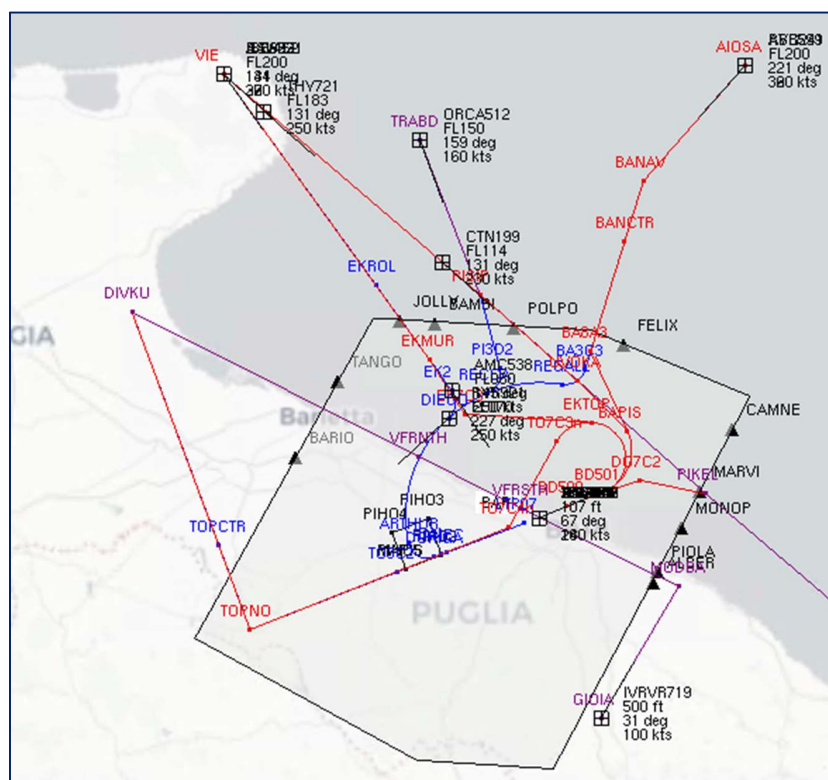


Fig. 7 Airways and SIDs/STARs used for operations in airspace classes D

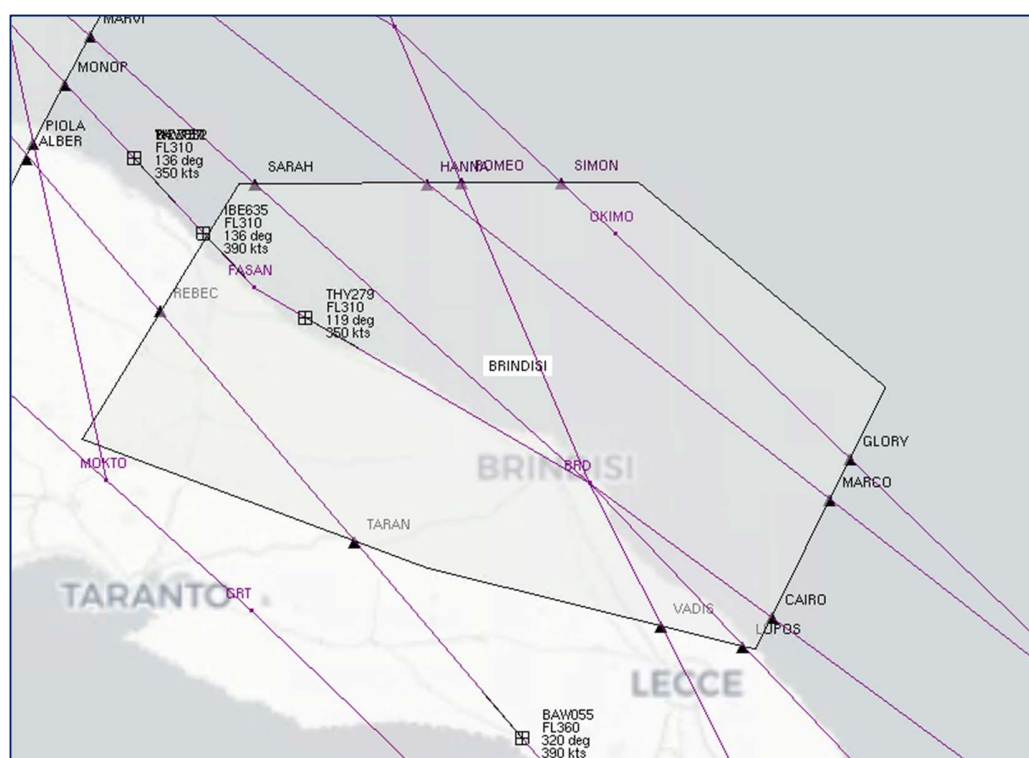


Fig. 8 Airways and SIDs/STARs used for operations in airspace classes A

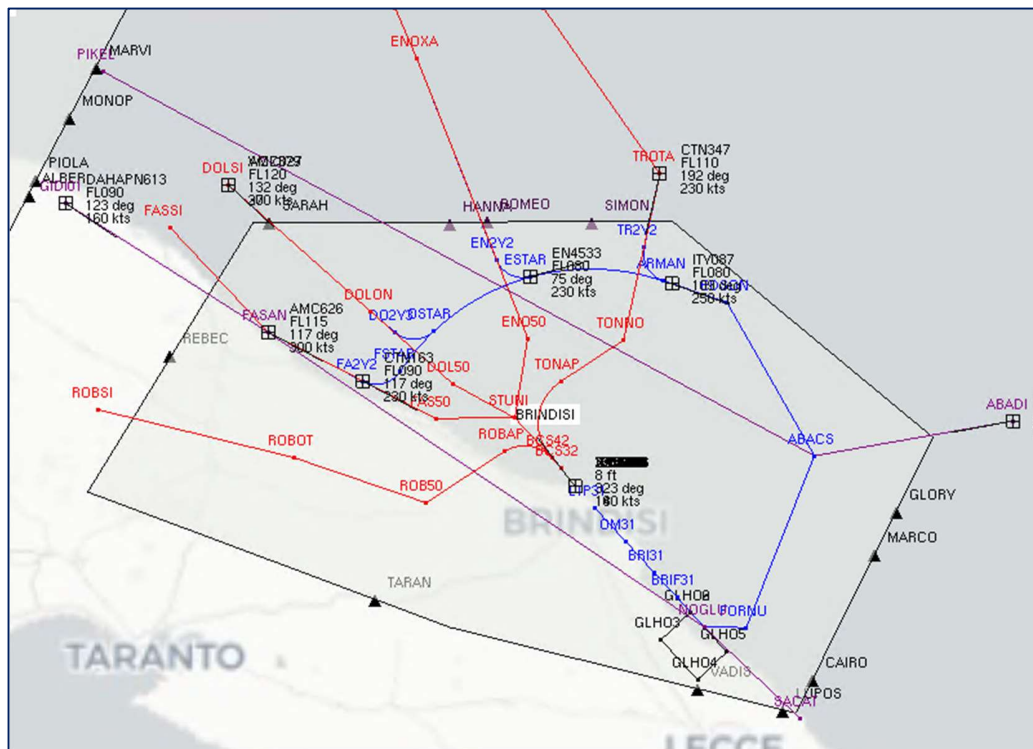


Fig. 9 Airways and SIDs/STARs used for operations in airspace classes E

For each test, the traffic actually used in simulations is generated randomly, at the traffic density level specified here below.

For any of the basic scenarios, traffic density has been set at one of two levels, a medium level of traffic, expected to generate a new item (aircraft) in simulation in a range between 4 and 5 minutes, and a high traffic simulation level, in which a new aircraft is added in the scenario every 3 minutes.

1. Intruder simulation

As anticipated previously, the intruder simulation has been finally implemented through the traffic simulator and pseudo-pilot control station. Four classes of aircraft intruder, making still reference to BADA classification, has been selected for intruders, as described in the table below.

Table 2 Intruder performance classes and characteristics

Config.	Equipment	Flight Rule	Airframe / BADA Class	Full Env Max FL	Full Env Speed (Kts)
A	Non-Coop, No Transp, No ACAS	VFR	Piston VLA, small GA / B1	185	70-220
B	Mode ES (ADSB-OUT), No ACAS	VFR	Turboprop GA up to 5 seats /B2	285	110-300
C	Mode ES (ADSB-OUT), ACAS Equipped	IFR	Turboprop GA up to 5 seats /B2	285	110-300
D	Mode ES (ADSB-OUT), ACAS equipped	IFR	GA Business jet up to 15-20 seats /B4	415	120-460

By referring this table, in the A configuration the aircraft is displayed on the Controller Working Position (CWP) just by a cross sign related to its horizontal (LAT-LON) position, with no other information reported: it is intended to reproduce the Primary Surveillance Radar (PSR) behavior.

In all other cases the aircraft provides to ATCO on the CWP-HMI the (LAT-LON) position by a filled square and the ICAO ID, and the altitude (ft or FL), the speed (m/sec) and the track (deg vs geographical North). It is also possible

to degrade the ADSB signal to only a Secondary Surveillance Radar (SSR) behavior (Transponder Mode A-C), in which just the (baro-)altitude is provided together with (LAT-LON) position.

As emerge from the Table 2, the IFR simulated intruder are equipped with ACAS, namely the TCAS-II implementation developed by EUROCONTROL and provided to CIRA in the form of a .dll executable SW function. The implementation allows both to be run with other non-ACAS equipped aircraft, and with TCAS-II equipped aircraft. In this case, in addition to normal "Traffic alert", also a "Resolution Advisory" is provided to both the two cooperating aircraft, providing each the resolution vertical maneuver. Also, the aural alert is provided other the visual one.

The TACS-II implementation should result in a cooperative operation with EUDAAS. In the figure below, the Intruder "IZAPN28" is suggested to climb at a rate of 1500ft/min in order to resolve the Collision risk with CIRA101 which is the CIRA-RPAS equipped EUDAAS system.

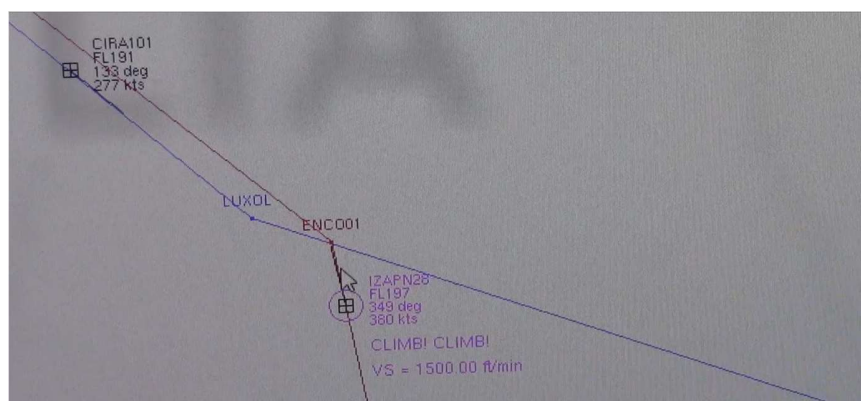


Fig. 10 Example of TCAS-II-RA activation on Intruder, as taken from the simulator during run.

All commands as applicable to traffic aircraft are also utilizable to manage Intruder flight, including emergency conditions coding.

For the execution of the CIRA RTS campaign the Apulia airspace has been considered, specifically using two simulated sectors, BRINDISI-CTR and BARI-CTR, around two airports, BARI-PALESE, BRINDISI-CASALE, all lying in the Brindisi Flight Information Region.

For the scope of the present project and the RTS test campaign, the sector BRINDISI-CTR, will be considered as Class E airspace from 1500ft AGL to FL115, will be considered as Class A from FL115 to FL195 and will be classified as Class C above FL195 till FL460, see Fig. 11. The sector BARI-CTR, will be considered as Class D airspace vertically extending from 1500ft AGL to FL195, and it is classified as Class C above FL195, see Fig. 11.

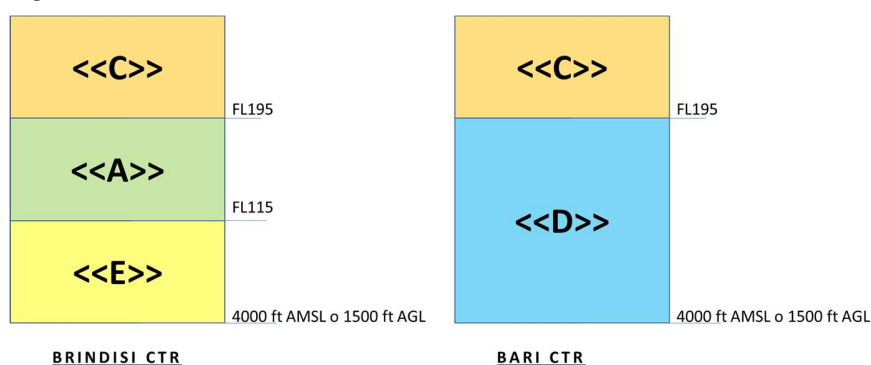


Fig. 11 Vertical Airspace classification over Brindisi and Bari CTR as assumed for RTS Tests

The RTS campaign has been organized by planning a number of Simulation Runs. Due to RPAS simulator mode of operations, each simulated flight needs to start from runways; two airports have been selected for take-off, one for the TUAV model, the Gioia del Colle airport and, for the MALE RPAS model, the Amendola airfield has been used. The scenario includes also some reserved areas for RPAS operations, in order to delineate more realistic kind of

missions: the existing reserved areas R315-R316-R317 around Grottaglie airport and the area R303A close to the Bari-Palese airport has been also included as characteristics of the scenario (see Fig. 12)

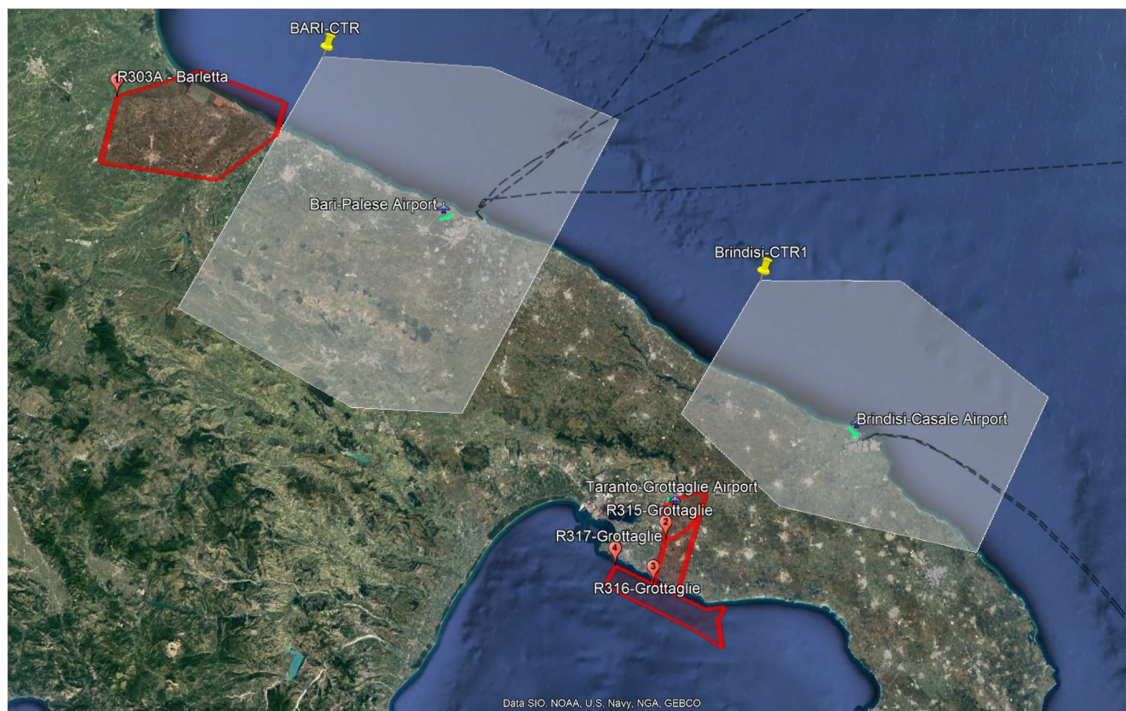


Fig. 12 CTR's and Reserved areas of operations implemented in the RTS facility

This opportunity emerged from preliminary evaluation held with professional ATCOs, that highlighted as relevant for the evaluation of DAA functionalities and, mainly, its effect on the interactions between RP and ATCO could be meaningful when an RPAS leaves a reserved area to enter the civilian unrestricted airspace.

2. Encounter Conditions

Two main reference sources have been used to derive the Encounter Conditions to be tested. The first one, the EUDAAS internal document Verification & Validation plan, for operations in airspace classes A and C, the second one, the RTCA MOPS for DAA system [4], for the operations and encounters in airspace classes D-E.

The encounter conditions will be specified by means of quantitative key parameters describing:

- the encounter geometry, both in the vertical and horizontal plane
- the intruder and ownship RPAS vertical and horizontal speed.

The selection of the encounter types under test has been done by means of joint work between the system requirement owner and the RTS facility owner, where the latest is in particular responsible to verify the compatibility of the proposed encounter conditions with the selected reference scenario configuration and with the RTS facility capabilities.

Finally, possible test conditions, including related encounter geometries, have been selected. An example of encounter and the related test conditions are described in the Table 3 and Fig. 13.

Table 3 Encounter synthetic description

Identifier	RTS.N01
Title	IFR/IFR encounter in class E with an overtaking intruder during RPAS level flight transit operation
Airspace Class	E
Intruders	Medium turboprop/jet aircraft (B3-B4). Cooperative with ADSB-OUT and transponder. ACAS-equipped
Encounter type	Traffic overtaking at same flight level

Description	Level flight point to point transit at <i>TBD</i> ft AGL on a fixed course. A business jet begins to overtake the RPAS on the same airways.
Operational Conditions	High Level surrounding traffic. Nominal conditions-Loss of C2L-DAA False Alarm

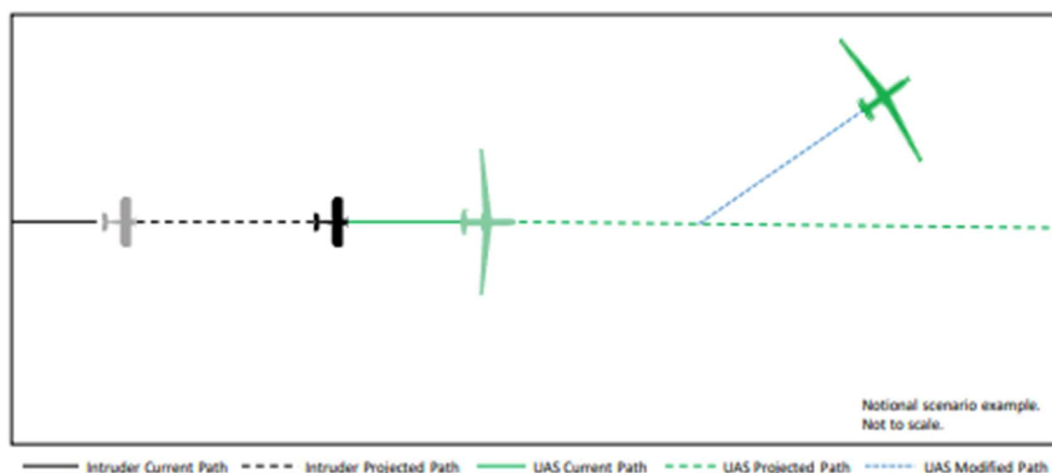


Fig. 13 Encounter with intruder overtaking the RPA

As concluding clarification of the scenarios implemented, Fig. 14 shows the Controller Working Position HMI for a specific case in which the equipped EUDAAS RPAS (CIRA101), the intruder (ITY325) and the surrounding traffic on Brindisi Casale airport flying the related STARs/SIDs, are displayed, as they have been actually operated and managed by the ATCO during a RTS test.

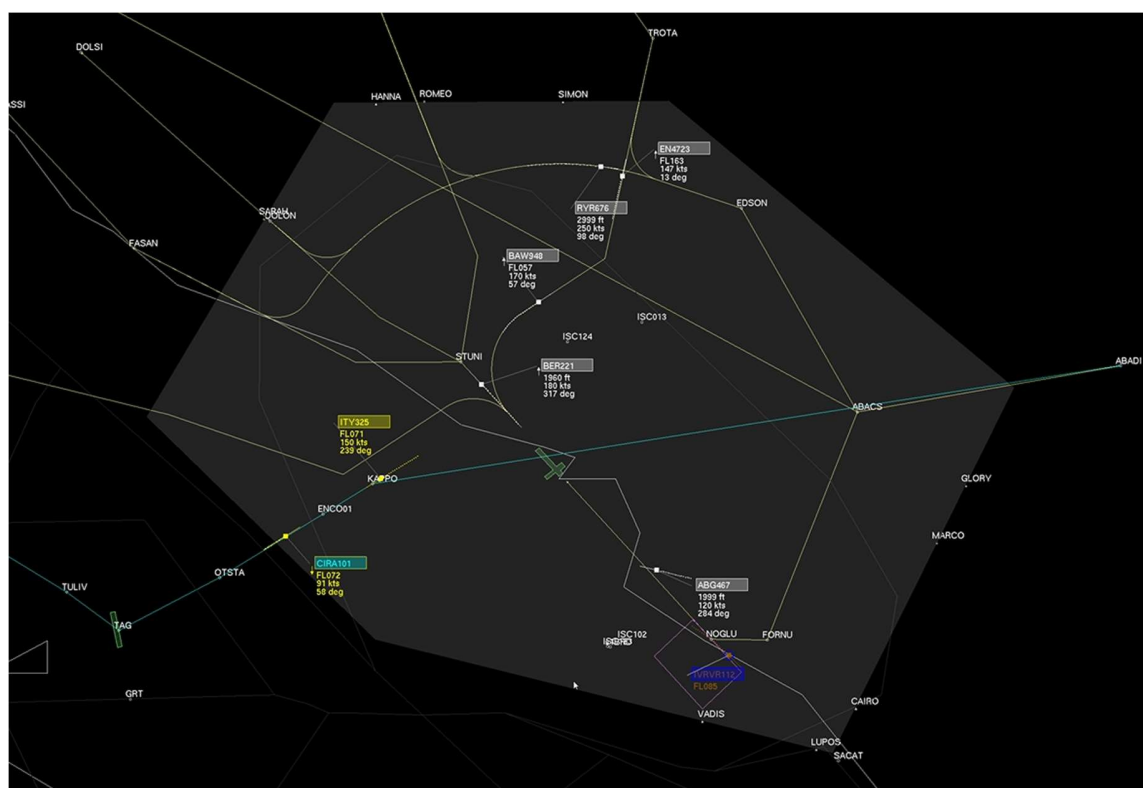


Fig. 14 ATCO HMI during TUAV test on encounter ENCE09.

IV. Human Factors Assessment

A. Background and validation objectives

The EUDAAS concept is derived from the MIDCAS (Mid-Air Collision Avoidance System) project, although it differs by several aspects (e.g., using suggestive guidance bands rather than an explicit maneuver suggestion for RWC). During the MIDCAS real time simulation campaigns, Alfredson et al [5][6] assessed the workload and the situation awareness of the remote pilots and air traffic controllers in a realistic air traffic environment. Overall, workload and situation awareness were deemed acceptable during these simulation campaigns. However, the following specific issues were highlighted and form a background for the EUDAAS RTS validation objectives:

- Awareness of the intruder relative position (especially during encounters involving several intruders and zooming) and equipage (e.g. coop. vs non-cooperative, transponder equipped or not).
- Acceptability of the guidance and avoidance maneuvers (recommendations for one single maneuver with a specified end point, limited number of dimensions changes, stable and consistent maneuver type);

As explained in section II, the Human Factors (HF) assessment to be performed during the EUDAAS real time simulation campaign is primarily addressing the acceptability of the DAAS and of the associated procedures by the remote pilot and the air traffic controller.

B. System concept

The EUDAAS concept proposed for HF assessment provides the following main capabilities:

- Traffic Awareness (TrAw) provides information that is relevant for DAA regarding other traffic and the own aircraft state to the RP. Surrounding traffic is monitored and its relative position and its velocity are indicated to the RP by means of a Human-Machine Interface hereafter called the DAAS display.
- Remain Well Clear monitors surrounding traffic and, if necessary, calculates and displays RWC suggestions to the Remote Pilot. The capability detects RWC conflicts, i.e. situations which are predicted to result in a LWC (Loss of Well Clear). When RWC conflicts occur, the capability raises an RWC alert to the RP. All intruders who are in conflict are indicated to the RP through the HMI
- Collision Avoidance (CA) monitors surrounding traffic, assists the RP in detecting collision threats and provides CA resolution alert and associated guidance for collision avoidance maneuvers. All intruders who pose collision threats are indicated to the RP through the HMI. If no timely response is undertaken by the RP, a CA maneuver is executed automatically.

A key feature of the DAAS concept is the suggestive RWC guidance, which can be either preventive (showing directions which would cause a RWC conflict to occur) or corrective (showing directions which would allow to solve a predicted RWC conflict) (Fig. 15). This guidance takes the form of guidance bands in the three possible maneuver directions (heading, altitude or speed) with a color coding intended to convey a sense of urgency to the RP (e.g. yellow for caution, orange for warning).

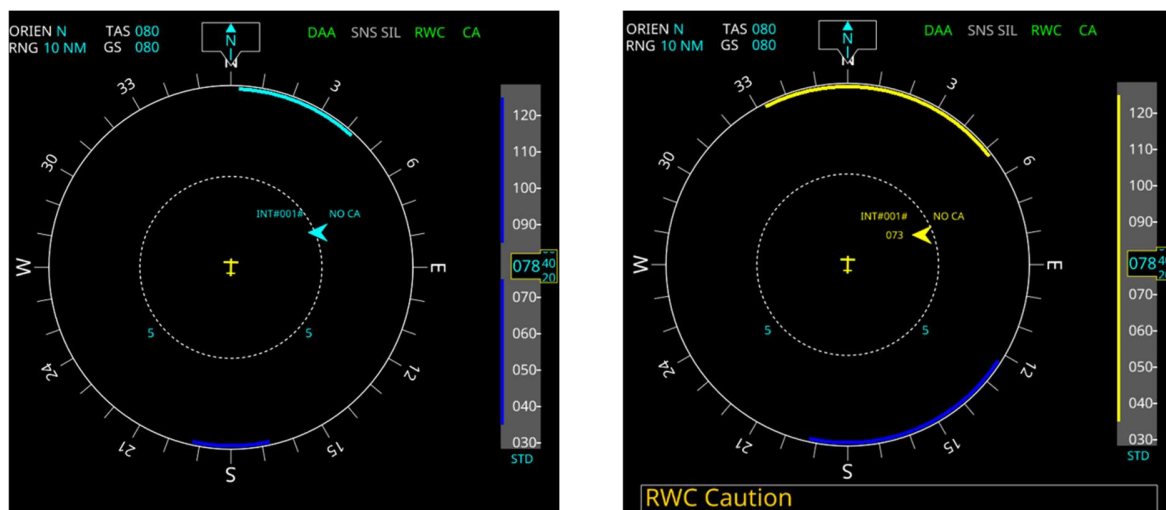


Fig. 15 Example illustrations of RWC Guidance: Preventive (left) and Corrective (right).

Note that the maneuver execution is not automatic. Instead, it is the sole responsibility of the RP to decide whether or not a maneuver should be performed, to choose the maneuver based on the available guidance and eventually to

execute it. Also, when relevant the RP has to coordinate with the ATCO before executing a maneuver. Note that this concept aims to be compliant with the minimum standard performance as set in ED-271A [7]. This concept obviously has human factors implications, which are the main focus of the HF assessment.

C. Participants

Two individual Remote Pilots and two individual ATCOs participate to the tests. All participants are highly experienced, both in military and civil aviation:

- The RPs both are flight test pilots with more than 5000 flight hours on many aircraft types, including rotorcraft and RPAS. They are used to TCAS. One of them also tested the MIDCAS system.
- The ATCOs both have long experience in civil and military Air Traffic Management, as controllers at various positions. They also have experience in the control of large military RPAS.

D. Procedures

The complete set of procedures to be followed by the Remote Pilot when using the DAAS are described in the DAAS RP Manual, which has been provided to the participants before the simulation campaign. The procedures describe in particular the reactions expected from the RP for each phase of a traffic encounter (TrAw, RWC, CA, Clear of conflict), depending on the airspace class and on the intruder equipment (cooperative or not). The procedures in case of non-nominal conditions are also provided: degraded VHF communication, loss of the Command and Control (C2) datalink, high latency C2 datalink.

The applicable procedures for use of the DAAS in nominal conditions are summarized below:

- The RP shall continuously monitor the DAAS display to maintain situational awareness with respect to surrounding traffic, supported by the RWC and CA capabilities;
- The RP shall not undertake manoeuvres based solely on the TrAw capability, unless it is supported by an amended clearance from the ATCO (when applicable);
- In controlled airspace, when an RWC alert ("Conflict, Conflict") is issued, the RP shall assess the traffic situation upon the DAAS display and, taking into account the suggested resolutions, request an amended clearance from the ATCO. After and if receiving an updated clearance, the RP shall implement the cleared maneuver to resolve the detected conflict;
- The RP shall monitor the evolving situation, and, when the TrAw and RWC guidance indicates it is safe to resume normal mission operations, request an amended clearance from the ATCO in order to do so.
- If a CA alert ("Avoid, Avoid") is issued, the RP shall immediately assess the traffic situation upon the DAAS display and decide to "Accept" or "Abort" the proposed CA maneuver;
- The RP shall monitor the execution of the CA maneuver and verify that the collision hazard has actually been avoided when the aural annunciation "Clear of conflict" is provided;
- In controlled airspace, the RP shall then communicate to the ATCO, when workload permits, that a CA resolution occurred.

Note that for this simulation trials the RPs only add to fly and to navigate the RPAS along a predefined flight plan, to manage its control systems, to communicate with the ATCO following the normal procedures and of course to use the DAAS as expected based on the DAAS RP manual.

E. Measures

Given the small sample of remote pilots and air traffic controllers available for the RTS-ATCO campaign, as well as the large number of simulations and criteria to be considered to fulfill the validation objectives, it was favored to perform a qualitative evaluation of the RP and ATCO interactions with the DAAS. Thus, the assessment is based on a set of material collected during the RTS-ATCO campaign which, combined together, allow to address the various facets of the acceptability of the DAA system. This material comprises:

1. Observer log

Notes are taken on the fly by the observers at the RP and the ATCO working positions. These notes are collected and used as support material to orient the debriefings, as well as for reporting purposes after each simulation session.

2. Instantaneous assessment of Workload

A verbal self-assessment of workload may be requested by the observers during the runs, especially at given occasions where the task load is expected to change (e.g. at intruder detection, after maneuver execution and after conflict resolution). Care is taken not to disturb the task at hand. Also, voluntary verbal self-assessment of workload is encouraged, when the actor feels it can be worth noted. The Modified Cooper-Harper rating scale was found as best

suitied for the workload assessment, as the flight test pilots involved in the tests are especially acquainted with this type of questionnaire.

3. Verbal reporting (when deemed informative)

"Loud thinking" is encouraged when delivering the simulation instructions to the participants and is collected through the audio recording of the simulation runs. These online comments from the participants are very useful to support the observer log and the debriefing.

4. Video / audio recording

Audio and video recordings provided by the RTS facility as well as the DAAS HMI video capture are used for further discussion and illustration when a point of interest is identified during the run.

5. Acceptability and usability questionnaires

An acceptability questionnaire derived from the Controller Acceptability Rating Scale (CARS) [8] is filled by the pilots and the ATCOs as far as feasible after each test (Fig. 16). In addition, a usability questionnaire derived from the System Usability Scale [9] is used at the end of each run to get an assessment of the usability of the DAAS from both the RP (Fig. 17) and the ATCO (where "control the RPAS" is employed rather than "use the system").

6. Debriefing

Semi directed interviews have been conducted along with the debriefing of each test. They aimed to highlight the HF issues encountered during the runs, and further discussions with the participants helped to explain their causes, their possible impact on the operation and some possible solutions for their mitigation.

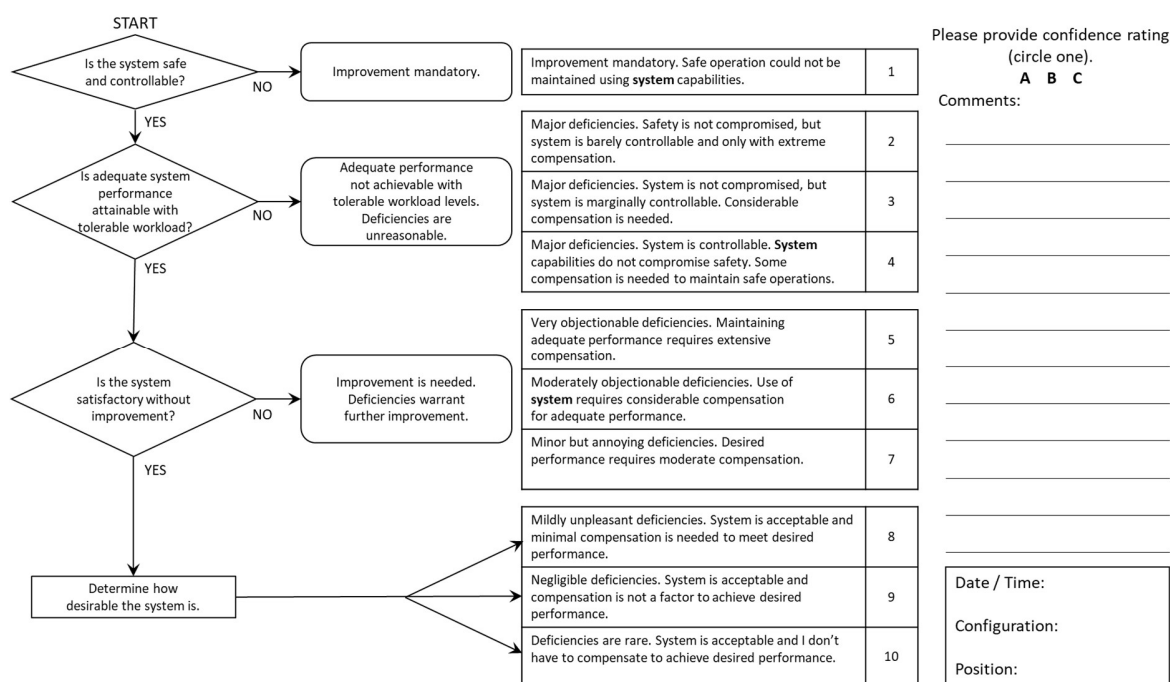


Fig. 16. Acceptability questionnaire for the RP.

		Strongly disagree				Strongly agree
#	Item	1	2	3	4	5
1	I think that I would like to use this system frequently.					
2	I found the system unnecessarily complex.					
3	I thought the system was easy to use.					
4	I think that I would need the support of a technical person to be able to use this system.					
5	I found the various functions in this system were well integrated.					
6	I thought there was too much inconsistency in this system.					
7	I would imagine that most people would learn to use this system very quickly.					
8	I found the system very cumbersome to use.					
9	I felt very confident using the system.					
10	I needed to learn a lot of things before I could get going with this system.					

Fig. 17. Usability questionnaire for the RP.

V. RTS test campaign execution

This section describes the actual execution of the real-time simulations, as carried out at CIRA premises, involving Remote Pilots and Air Traffic Controllers in the validation activities of the EUDAAS system.

Furthermore, this section describes the kind of data collected in each test and available to the project partners.

A. Test Campaign Organization

1. Planning

One test day was organized prior to the RTS campaign. Then the RTS campaign was organized in two sessions of one full week for each session. A few weeks were available in between, in order to allow for preliminary analyses and some minor adjustments.

2. Team role and responsibilities

A number of professionals have been involved in the RTS campaign at CIRA, namely:

- 2 Remote Pilots;
- 2 Area/APP Radar ATCO;
- Manned IFR intruder pseudo-pilot;
- Traffic pseudo-pilot.

In addition, in order to support the correct RTS campaign execution and to collect relevant performance data, both from system and humans involved in the validation test, the following experts attended to the test campaign:

- 5 test engineers, that is typically CIRA engineers who assure the correct plan and execution of tests, and
- 2 Human Factors experts who are responsible for the human performance measurements and coordinate the evaluation of results.

B. Collected and available data

During the test many types of data have been collected and have been stored. These are available to the project team, for any analyses by the partners.

For each test, in particular, CIRA stored the following kind set of data:

a. TIME Flight ID Position Inertial speed attitude angles ModeACode

for all the traffic scenario aircraft, on the Traffic Simulator. In addition, the time and number of PPT (push-to-talk) activation by the traffic pseudopilot have been recorded for possible measures of workload.

b. TIME Flight ID Position Inertial speed attitude angles ModeACode TCAS messages

for the intruder, on the Intruder simulator. Other than the variables recorded for traffic aircraft, also three variables related to the TCAS-II activation, when available, have been recorded for the Intruder. Again, the PPT use has been recorded for workload derivation.

c. TIME ENTITY Position Inertial speed BAROALT(m)

Have been recorded on the ATCO-CWP, for all the entities included in the test, that is the RPAS (TUAV and MALE), the intruder and the opportunity traffic aircraft.

Also, for the ATCO, the frequency of use of the PPT is recorded.

d. The complete set of Position, Velocity and Time data and the housekeeping data of all the on-board systems are recorded for the RPAS and on a Simulation Control Facility.

The data are recorded in a proprietary format and need to be processed before to be analyzed by personnel external to CIRA. The data are not "reserved" and can be made available to the project partners.

In addition to ASCII and raw data, a complete set of audio-video recording files are available. Audio-video recordings have been taken on all the following components of the overall RTS facility used in the EUDAAS RTS test campaign:

1. Controller Working Position, on which the CWP-HMI (radar screen) of the ATCO activities and communications have been recorded;
2. Remote Pilot position, on which the RP-HMI (Remote pilot Control Station) of the remote pilot activities and communications have been recorded;
3. The DAA system interface, on which the DAA system behavior and the RP activities on it have been recorded.

e. The complete set of data in input and output to the DAA model are recorded on the network of all the on-board systems are recorded for the RPAS simulation facility and on a Simulation Control Facility.

C. Description of the Test Report Information

The following table describes the information reported for each test performed during the two simulation sessions.

Table 4 Test report table description

Test ID	Identifier of the test as reported in the Test Plan
Test Info	Number of the RTS session; Data of execution of the test; Incremental number of the test made in the day; Number of encounters inside the current test.
Encounter Description	Synthetic description of the encounter; Airspace class in which the test is executed; TMA/No TMA
Off-Nominal Events Injection	Description of the off-nominal events that have been injected into the test scenario (e.g. C2 link loss, C2 link delay, etc). N/A if no off-nominal events have been injected during the test.
UTC Time at start of the DAA HMI Video	UTC time associated to the start of the DAA HMI video.
Ownship Type	Ownship type: TUAV o MALE; Mean speed and Flight Level of the RPAS during the encounter.
Intruder Type	Intruder type: IFR/VFR; Intruder equipment; Mean speed of the intruder during the encounter.
Traffic Type	Traffic density: High, Medium, Low.
Alert Activation	Sequence of alerts activation during the encounter together with the DAA HMI video's time instants.
RP Manoeuvre	Description of the RP/CA manoeuvre during the encounter, if any, together with the DAA HMI video's time instants. Here is also reported if any clearance has been asked to the ATCO before manoeuvring.
Minimum horizontal distance and related relative altitude	Minimum horizontal distance and related vertical distance during the encounter. The values are computed using the real aircraft data (not measured).
Validated Objectives	Validation Objectives covered by the test.

Execution Result	Test correctly executed/Test partially executed/Test failed. Indicates if the execution of test (not the DAA results for which a deep analysis is needed) was correct/partially correct/not correct.
Deviations from the Test Plan	Description of possible deviations of the executed test with respect to what effectively planned.
Notes	Description of problems or unexpected events detected during the test.

D. RTS execution and example test cases

A total of 39 tests were conducted during the RTS campaign, including repeated executions of selected scenarios to ensure consistency and result validation. In addition to the planned test cases, a small number of unplanned scenarios were executed based on inputs from RP and ATCO, focusing on additional operational conditions deemed relevant. A statistical overview of the test distribution is summarized below and depicted in the Fig. 18.

The tests were distributed across four airspace classes (D, E, C, and A). Regarding the intruder aircraft, the scenarios were almost evenly divided between IFR and VFR operations, with 19 and 20 tests respectively.

Regarding operational conditions, 9 tests were conducted under nominal conditions, while 30 tests included non-nominal or system-degraded conditions, such as:

- RPAS high-latency C2 links,
- Temporary C2 link loss,
- VHF communication degradation or loss.

To simulate these non-nominal conditions, a specific functionality was implemented in the simulation environment, allowing for dynamic manipulation of the communication channel among simulation entities. This feature enabled realistic emulation of communication delays, partial losses, or disturbances with varying readability levels (from 5/5 to 1/5), thus replicating degraded operational environments.

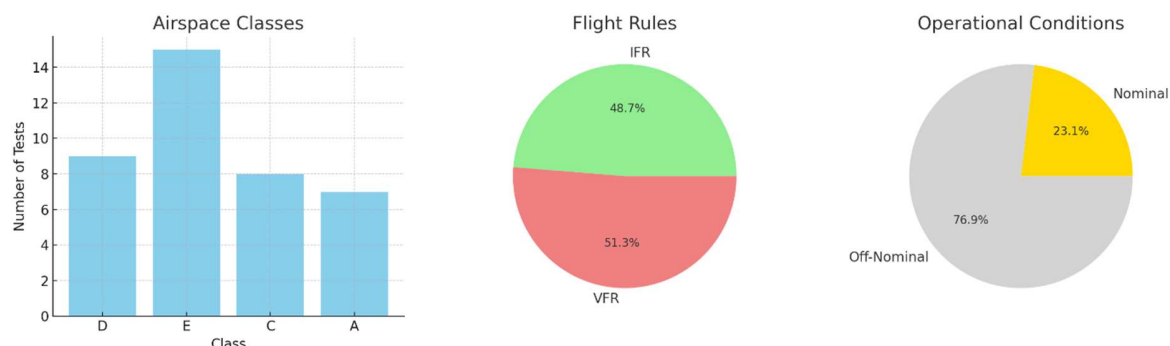


Fig. 18 Statistical distribution of RTS campaign tests by airspace class (left), intruder flight rules (center), and operational conditions (right).

Following the detailed analysis of the results for two examples of alerting sequence on the DAA HMI are reported. The first, characterized by a co-altitude IFR/IFR conflict within airspace Class C, outside of Terminal Maneuvering Area (TMA) boundaries. The ownship, a MALE RPAS flying at FL200 with a true airspeed (TAS) of 284 knots, faced an intruder aircraft operating under IFR and broadcasting via ADS-B OUT (Mode ES) sensor, at a TAS of 380 knots, in a high-density traffic environment. The scenario included an off-nominal condition consisting of degraded VHF communications during the encounter, designed to challenge the situational awareness and response capability of the RP.

The alerting sequence on the DAA HMI began with a Caution alert at 15 minutes 30s (see Fig. 19), followed by an Advisory alert at 16 minutes 04s (Fig. 19) after the RP maneuver, and cleared to No Alert at 16 minutes 13s. In response to the caution condition, the RP contacted ATCO and requested clearance for a left maneuver, initiated at 15 minutes 56s, as suggested by the RWC function on-board. Despite the degraded voice communication, the RP successfully executed the maneuver in coordination with air traffic control.

The minimum horizontal distance recorded during the encounter was 1,88 nm on the horizontal, with a relative vertical separation of 203 feet. These values indicate that although the conflict involved minimal vertical separation, sufficient horizontal margins were maintained throughout the encounter keeping the well-clear conditions.



Fig. 19 DAA HMI alert activation during RWC function test

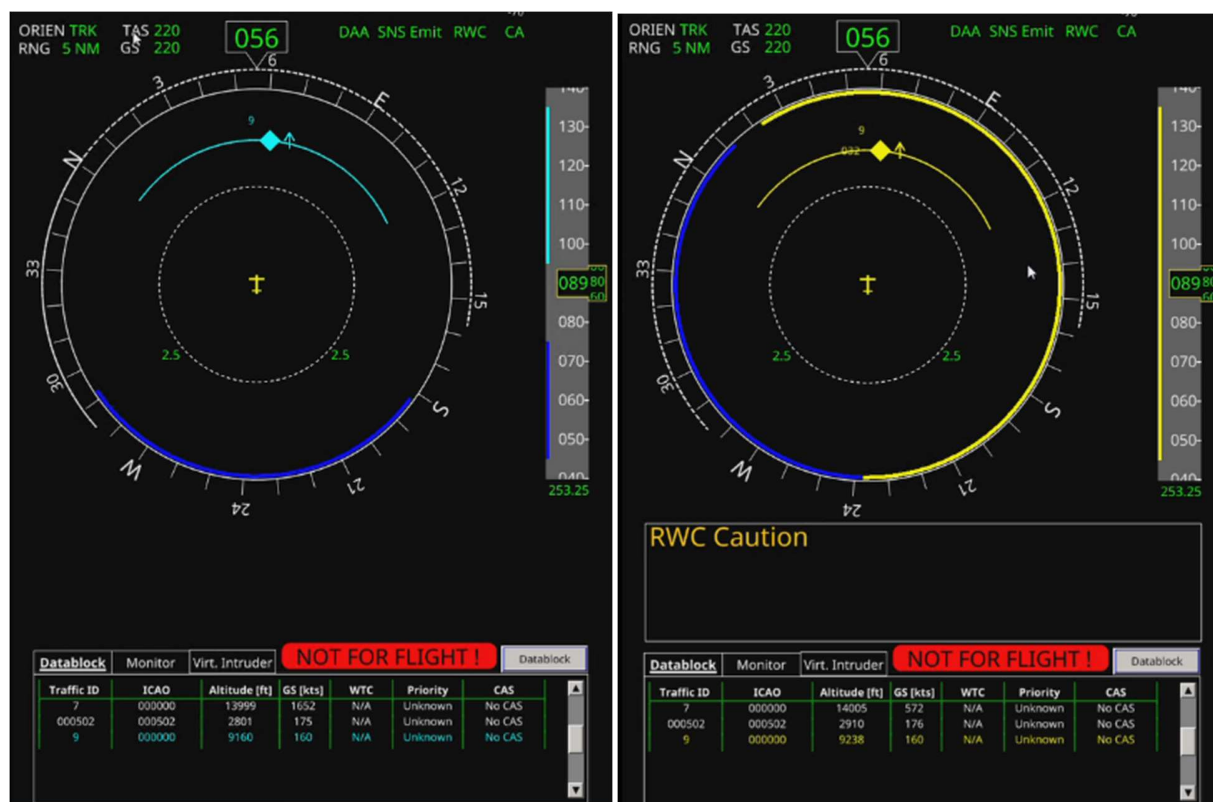


Fig. 20 Example advisory and RWC caution alerts



Fig. 21 DAA HMI alert activation during CA function test

In a second example scenario a head-on encounter has been simulated between a MALE RPAS and a non-cooperative VFR intruder during a climb phase transit in Class E airspace, within a Terminal Maneuvering Area (TMA). The ownship was climbing at FL90 with a TAS of 220 knots, while the intruder, a non-cooperative VFR aircraft, approached at 160 knots in a medium-density traffic environment. No off-nominal events were injected, allowing the test to focus on the behavior and effectiveness of the Detect and Avoid (DAA) system under nominal conditions.

The encounter triggered a progressive sequence of DAA alerts based on the detection of the intruder by means of the radar on board the ownship. First an Advisory alert at 12:07 (Fig. 20) was issued, escalating to Caution at 12:10 (Fig. 20), followed by LOW at 12:12 (Fig. 21), and culminating in a Collision Avoidance (CA) alert at 12:28 (Fig. 21). The Collision Avoidance maneuver, consisting of a right turn, was triggered automatically by the system at 12:32.

The minimum recorded distance during the test were 0.2 NM horizontally and 295 feet vertically, demonstrating the effectiveness of the CA system avoiding the breach of the Near Mid-Air Collision (NMAC) volume.

VI. Results of the Human Factors Assessment

A. Limitations

Despite the satisfying test cases presented previously, the results highlight important deficiencies in the current implementation of the DAAS, impacting the RP attention and workload, the RP-ATCO coordination and finally the RP performance in terms of decision making and action. These deficiencies mostly resulted from a lack of integration verification due to time pressure in the project development. They hinder the overall assessment of the system. More insights for proper operational use of the system would require new tests once these deficiencies are fixed.

B. Main findings

Notwithstanding these limitations, key attention points as well as recommendations for improvements in the design of the DAAS can be provided, based on the lessons learned during this real-time simulation campaign. These points are presented and discussed in the next sections, regarding each capability of the DAAS.

1. Traffic Awareness

In general, the traffic situation is assessed as correctly displayed. The DAAS provides information consistent with existing traffic displays and in accordance with current procedures and rules.

In particular the non-cooperative traffic can be clearly differentiated from the cooperative traffic (use of non-directional vs directional traffic symbols).

However, the clutter of the DAAS display was clearly identified as an issue.

In particular, traffic far above or far below the ownship was frequently displayed on the HMI, although it was no concern for the RP. The addition of such type of traffic makes it difficult to discriminate between relevant and irrelevant traffic, drowning out the useful piece of information. This issue was further aggravated by the fact that the relative altitude and vertical tendency of the traffic were not always displayed, depending on the quality of the relative vertical positioning. Enhanced filtering criteria should be developed in order to mitigate this issue.

Another issue was related to the scale management of the DAAS display. The scale/range management required two mouse clicks on the HMI, making it difficult to use in practice. This may affect the traffic awareness, as the RP will not change the scale of the display as they may wish. Later on, similar difficulties also appeared for the implementation of a maneuver. The proper integration of the HMI controls is a prerequisite for the ease of use of the DAAS. For instance, rotary knobs for the scale management as well as target altitude command would make the RP task easier.

2. Remain Well Clear

The RWC task is affected by the logic of the DAAS as well as the design of the DAAS display, including the related alerts and guidance.

When under ATC separation, the time available was generally short for the RP to complete the required tasks, including the coordination with the ATCO before an amended clearance can be obtained. It was specified that the DAA shall issue a RWC alert to the RP as soon as possible but no earlier than 75 seconds ahead of a predicted loss of well clear. In practice during the tests the RWC alert was frequently issued with much shorter times (depending on several parameters, and especially sensor limitations for the detection and tracking of non-cooperative traffic). Within the remaining time, the RP has to complete the following steps:

- First, the RP needs to decide of a possible resolution maneuver, in accordance with what is proposed by the DAAS. Yet, frequent changes in the direction, amplitude and nature of the guidance sectors made the decision difficult or impossible.

- Then, the RP has to request clearance. This can be made more or less difficult depending on the number of requests the ATCO is currently facing from other aircraft and on the quality of the VHF communication. On several occasions during the simulations the RP were unable to do so immediately because of dense traffic.
- After receiving the clearance from the ATCO, the RP needs to prepare and then to engage the maneuver. As mentioned previously poor integration of the autopilot commands can make this step cumbersome.

In its current implementation, the RWC guidance bands are presented with different colors (yellow, orange, blue, red) depending on their status (e.g. preventive vs corrective) and on the conflict severity level. Note that the changes of color attract the RP attention, while it should rather be directed to the sectors suggested to solve the conflict. The direction and the extent of the guidance bands may also change rapidly, as a function of the intruder tracking and resolution algorithms. As a result, the guidance bands were found as confusing and difficult to interpret within the remaining time. A recommendation was to further simplify the DAAS logic and the display in order to ease the interpretation in dynamic situations.

Another concern is the choice of the maneuver direction. In some encounters, the RWC guidance suggestions are both on horizontal and vertical plane, with yellow sectors covering the current heading and altitude. Following the RWC guidance indications and the recommended procedure the RPs always request clearance in only one dimension. It is important to insist in the DAAS RP manual that no mixed maneuver is expected; the RP indeed has to choose to maneuver either in the vertical or in the horizontal dimension. The RWC guidance algorithm should be consistent with this constraint.

3. *Collision Avoidance*

The concept of automatic execution of the CA maneuver was found worthwhile and acceptable.

However, during some tests, the timing and the execution of the maneuver were difficult to understand for the RP. For instance, the automatic CA maneuver was frequently initiated immediately. In that case, there was no possibility for the RP to anticipate its execution ("Accept" button). Note that the DAAS RP manual was lacking the explanation that CA activates automatically at the last point to go outside the CA volume, which means that there could be cases of late detection, when the RP does not have time to accept.

By design, once the conflict is solved the DAAS releases the RPAS flight with straight and level flight. The RP found the "Clear of conflict" annunciation acceptable, as well as the time countdown displayed before release, allowing to request a new clearance by the ATCO when applicable.

4. *DAAS alignment with platform performance*

The RTS trials involved two types of platform: MALE and tactical RPAS. The flight performances of these platforms differ substantially. The DAAS algorithm take the platform performance into account. However, the dynamics of an encounter may change radically depending on the ownship performance, and in some cases because of the slow speed of the tactical RPAS the loss of Well Clear can hardly be avoided. Also, the RWC vertical guidance indications quickly saturates in the tactical RPAS case, due to its limited rate of climb/descent.

The logic of the DAAS design must align directly to the performance of the platform in which it is installed, with the related RWC and CA timings adjusted appropriately and possibly with extra margins.

5. *Required skills and training*

After completing the simulation campaign, the RPs stated that the roles attributed to them and induced by the DAA task in controlled airspace may require an adaptation of skillsets and training.

Notably, several display elements or functionalities were not found straightforward to understand and use, especially when they differ from classical TCAS displays or CDTIs. Thus, RPs need extra training and time to adapt to them.

Likewise, compared to the procedures and safety rules currently followed by the RP and the ATCO, the DAAS induces avoidances within much shorter times and distances. This leads pilots without training regarding this system to exhibit, in their opinion, a much riskier behavior than usual. Consequently, habituation to the system performances and behavior is necessary.

On the ATCO side, throughout the whole evaluation it was found that interacting with the RPAS didn't change their usual procedures. Still one warning was raised regarding the habituation to the shorter distances to wait for before a RWC alert, as instructed for these particular tests. Indeed, the ATCO mentioned on several occasions that if they were to follow classical procedures with cooperative traffic, they would have anticipated the encounters a long time in advance and would have required the RPAS to maneuver sooner.

C. EUDAAS2 preliminary results

The EUDAAS project ended on 30 November 2024, but the activities continue under the EUDAAS2 project, which started on 1 December 2024.

Based on the main results and overall evaluation reported above in the first six months of the EUDAAS2 project, an updated version of the DAAS model has been tested. The main changes in this new version are:

- Several software bugs fixed
- Reduction of CDTI cluttering based on a relative altitude selectable by the RPA and removal of the uncertain band on the air traffic tracks
- Improvement of the accuracy and the stability of the air traffic tracks information
- A new logic for the RWC bands, i.e. added green color to highlight where to go, see Fig. 22
- The RWC and CA algorithms have been revised in order to meet the RP request for more acceptable maneuvers in the vertical plane.

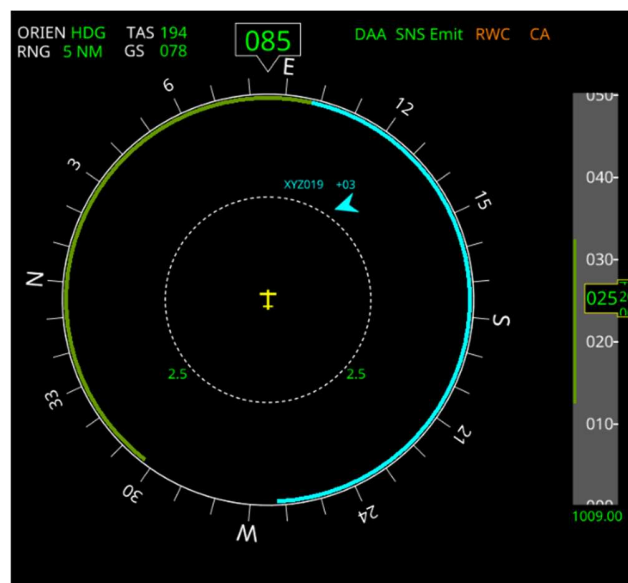


Fig. 22 DAA display, illustration of preventive with the green band

A preliminary simulation session with the RPs was performed the first week of June 2025. The RPs confirmed the improvement in the EUDAAS system. In particular, they noted a greater stability of the information provided and found the green band useful.

VII. Conclusions

This paper describes the real-time simulation campaign with remote pilot and controller in the loop carried out in support of the operational validation of the European Detect And Avoid System (EUDAAS) project.

In particular, in this work have been recalled: the validation objectives; the test cases and the scenario used; the architecture of the simulation environment; the methodology applied for the Human Factors Assessment and the procedure and the data collected during the RTS.

Finally, some examples tests have been analyzed and tests results have been reported.

Overall, the participants agreed that a DAAS is needed for the proper integration of RPAS in the air traffic system, which is indeed the aim of the DAAS under evaluation.

In the tested implementation, the acceptability and usability ratings of the DAAS by the RP were low, mainly because of the integration deficiencies which hinder a proper evaluation of the concept. On the opposite, ratings by the ATCO were good, as the details of the design did not impact the ATCO task compared to the RP task.

The RTS-ATCO simulation environment was generally found satisfactory for the purpose of the tests. The facility including the RP and ATCO working positions as well as the pseudo-pilot station allowed simulating realistic operational conditions, including non-nominal test cases.

This RTS-ATCO simulation campaign succeeds to highlight the potential benefits and practical issues in the use of the DAAS in realistic dynamic operational conditions, involving highly experienced participants. Several

recommendations were formulated as a result of the campaign, which will help refine the concept and the design of the future European Detect And Avoid System.

The work began in EUDAAS project is ongoing in EUDAAS2 project started on 1st December 2024, a preliminary RTS session based on updated version of the DAA system was performed in the first week of June 2025.

Acknowledgments

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